

SinhalaJournal-LLM

A Style-Adaptive Large Language Model Framework for Sinhala Journalism

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering
Project ID: R26-SE-037

Member 1 – Grammar Checker
Member 2 – Headline & Image Generator
Member 3 – News Summarizer
Member 4 – Sinhala Journalistic Style Rewriter

Research Team



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THE PROBLEM

- Lack of AI-powered editorial tools for Sinhala journalism
- Manual grammar correction, rewriting, headline generation, summarization
- Time-consuming and inconsistent workflow

USERS AND STAKEHOLDERS

Primary

- Busy Sinhala Journalists
- Sub Editors
- Digital Content Creators

Secondary

- Media Institutions
- Online news publishers
- Newspaper Agencies

Territory

The Sinhala-speaking public (approx. 21 million people, including 11.5 million active internet users)

PROPOSED SOLUTION – SINHALAJOURNAL-LLM

Unified AI newsroom assistant
Journalism-domain fine-tuned LLM backbone

Sinhala Grammer
Checker

Sinhala News
Summarizer

Sinhala Adaptive
Headline and Image
Generator

Sinhala Journalistic Style
Rewriter



Sinhala Grammar Checker

Problem

Sinhala digital news platforms lack an advanced grammar correction system

Research Gap

Current Sinhala grammar correction tools rely mainly on rule-based methods and do not use Sinhala Large Language Models.

Novelty

A journalism-focused Sinhala grammar system that combines a hybrid system

Objective

To develop a hybrid Sinhala grammar correction system using a fine-tuned Sinhala LLM integrated with rule-based validation.

Sinhala Grammar Checker

Sub-Objectives:

1. Prepare Sinhala sentences with grammar errors and their correct versions.
2. Fine-tune the SinLlama model for grammar correction.
3. Develop rule-based grammatical validation rules.
4. Integrate both approaches into a hybrid correction framework.
5. Evaluate performance using standard NLP metrics.

Contribution to the Overall Solution: Ensures accurate grammar across all SinhalaJournal-LLM outputs

Sinhala Grammar Checker

Research Methodology

Design Science Research approach within the NLP and AI domain, focusing on building and experimentally evaluating a hybrid grammar correction system

Latest Technologies Used

Fine-tuned
Sinhala Large
Language Model
(SinLlama)

LoRA (Low-Rank
Adaptation) for
efficient fine-
tuning

Transformer-
based
architecture

Rule-based
grammatical
validation
module

Data is collected from publicly available Sinhala news sources

Sinhala Grammar Checker

System

Input Article



Text Preprocessing



Rule-Based Grammar Validation



Neural Correction



Hybrid Integration



Final Corrected Output

wms wo mdi,g .sfhah

[wm\$ [wo] [mdi,g]sfhah]

Grammar Error → Subject–Verb mismatch

Generate correct verb

Combine rule + AI result

wms wo mdi,g .sfhuq

Evaluation

Precision, Recall,
F1-score, and
GLEU metrics

Validation

Human evaluation



Style-Adaptive Headline & Image Generator

Problem

Sinhala digital news platforms rely on manual headlines and images, causing inconsistency and slower publishing.

Research Gap

No AI system currently generates style-controlled Sinhala news headlines with aligned images, so newsrooms rely on manual headline writing and image selection.

Novelty

Uses a fine-tuned Sinhala LLM to generate style-based headlines and matching visual prompts.

Objective

Develop a Sinhala LLM-based system that generates style-adaptive headlines and semantically aligned visual prompts.

Style-Adaptive Headline & Image Generator

Sub-Objectives:

1. Build a Sinhala news dataset with style labels.
2. Fine-tune SinLlama for headline generation.
3. Implement style-controlled headline generation.
4. Extract key entities and keywords from headlines.
5. Generate aligned image prompts using multimodal AI APIs.

Contribution to the Overall Solution: Generates style-controlled Sinhala headlines and aligned visual prompts for newsroom content creation.

Style-Adaptive Headline & Image Generator

Research Methodology

Design Science Research approach focusing on developing and experimentally evaluating a style-controlled headline generation framework.

Latest Technologies Used

Fine-tuned
Sinhala Large
Language Model
(SinLlama)

LoRA (Low-Rank
Adaptation) for
efficient fine-
tuning

Transformer-
based
architecture

Open
Source API-
based
generation

Data collected from publicly available Sinhala digital news platforms.

Style-Adaptive Headline & Image Generator

System overview

Input Article



Text Preprocessing & Tokenization



Style Parameter Injection



SinLlama Headline Generation



Headline Optimization



Semantic Feature Extraction



Visual Prompt & Image Generation

කොළඹ නගරයේ නව බස් මාර්ගයක්
හඳුන්වා දීම

["කොළඹ", "නගරය", "නව", "බස්", "මාර්ගය"]

Style = Breaking News / Formal / Youth Style

කොළඹට නව බස් මාර්ගයක් හඳුන්වාදෙයි

කොළඹට නව බස් සේවාවක් – මාර්ගයක් අද
හඳුන්වාදෙයි

Entities → "කොළඹ", "බස් සේවාව"
Keywords → "නව මාර්ගය", "ප්‍රවාහන"

City bus traveling in Colombo urban road,
public transport news scene

- Other components' outputs validated before final response

Validation & Evaluation

ROUGE, BLEU, and human evaluation for relevance and style accuracy.

Output

- Style-controlled Sinhala headline
- Semantically aligned generated image



Sinhala News

Summarizer

PROBLEM

- Extractive methods copy sentences from the article
- Extracted sentences may break context or flow
- General LLMs may generate incorrect or hallucinated facts



RESEARCH GAP

- Existing Sinhala summarizers mainly use extractive approaches
- Generated summaries often copy sentences instead of rewriting meaningfully
- No summarization system designed for Sinhala journalism writing style

NOVELT

- Y**
- Generate summaries without copying sentences from the article
 - Preserve original news meaning and context
 - Maintain journalistic Sinhala writing style



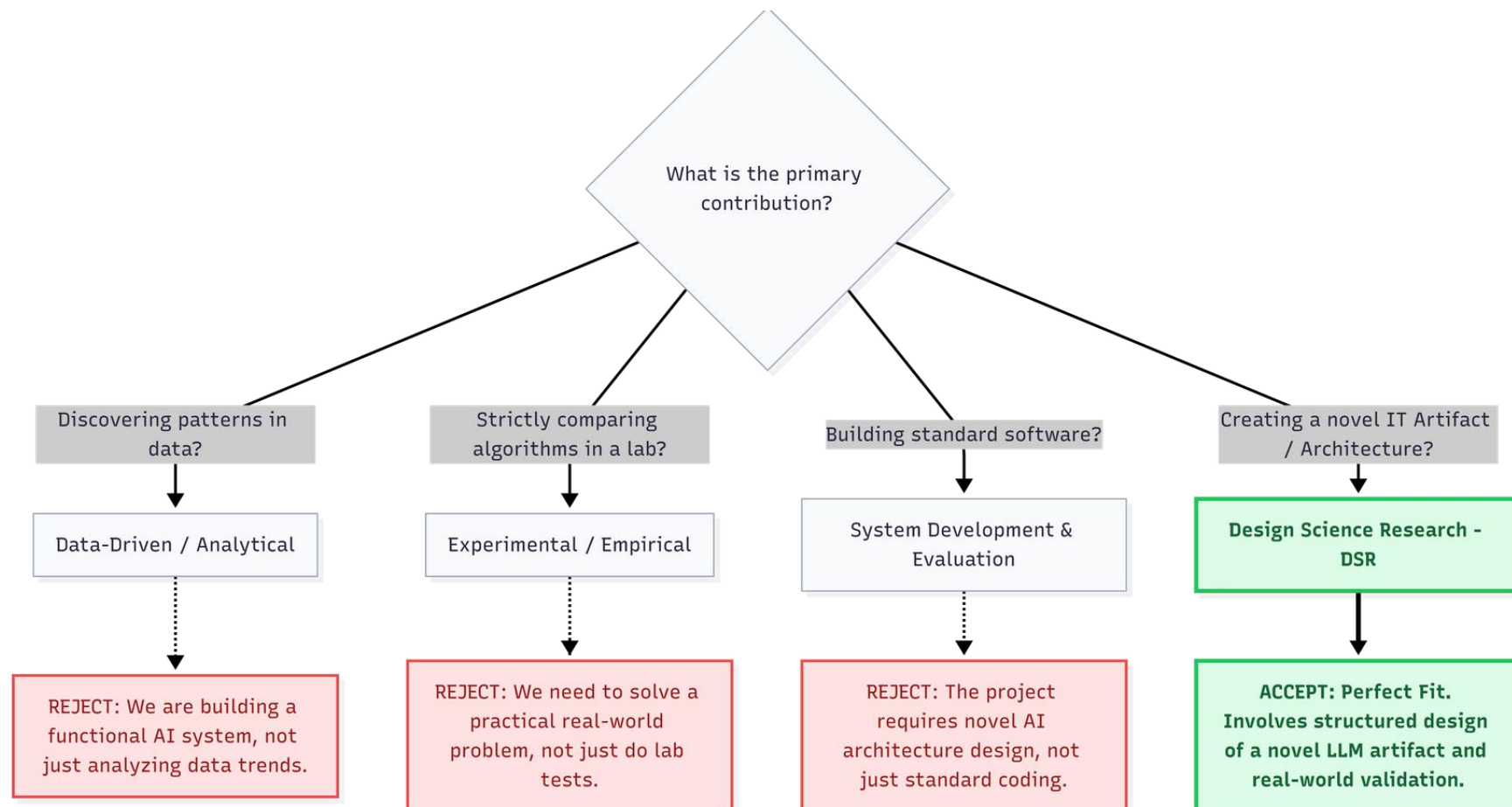
OBJECTIVES

- Build Sinhala news article dataset
- Design meaning-preserving summarization model
- Implement factual consistency checking
- Evaluate summaries against human references

RESEARCH METHODOLOGY

Design Science Research (DSR)

- Build real AI solution
- Solve newsroom workflow problem
- Evaluate with real news data



Sinhala News

Summarizer

TECHNOLOGIES USED

Base Model

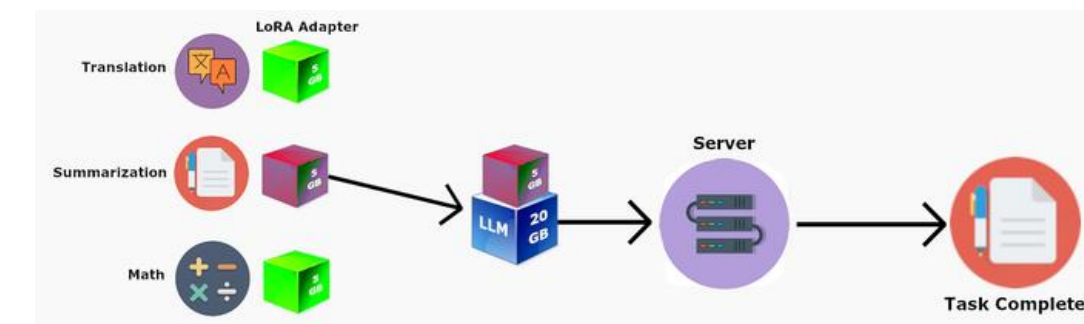
SinLlama - Sinhala LLM



Efficient Training

LoRA (PEFT)

Faster Training, Lower Compute

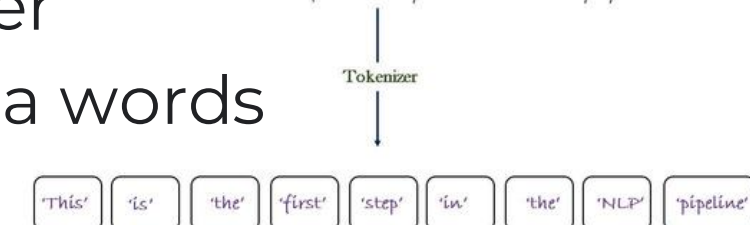


Language Processing

Byte-Level BPE Tokenizer

Handles complex Sinhala words

"This is the first step in the NLP pipeline"



QUANTITATIVE TESTING

We will use ROUGE metrics. This formula checks how many words from our AI summary match a summary written by a real human.

QUALITATIVE TESTING

Since ROUGE only checks exact words, we will also use a bigger AI (like GPT-4) as a judge to check if the meaning and facts are 100% correct.

DATA COLLECTION

We collected training data by scraping public news websites like Ada, Hirunews, Mawbima and Lankadeepa.

SECURITY & ETHICS

Journalists work with top-secret news. Our system uses temporary memory (ephemeral processing). This means the news is deleted immediately after the summary is made, keeping their data safe.

SYSTEM OVERVIEW

Context Window Manager

- Splits long Sinhala articles
- Fits LLM token limits

Abstractive Generation Node

- SinLlama + LoRA model
- Generates human-like summary

Factual Consistency Filter

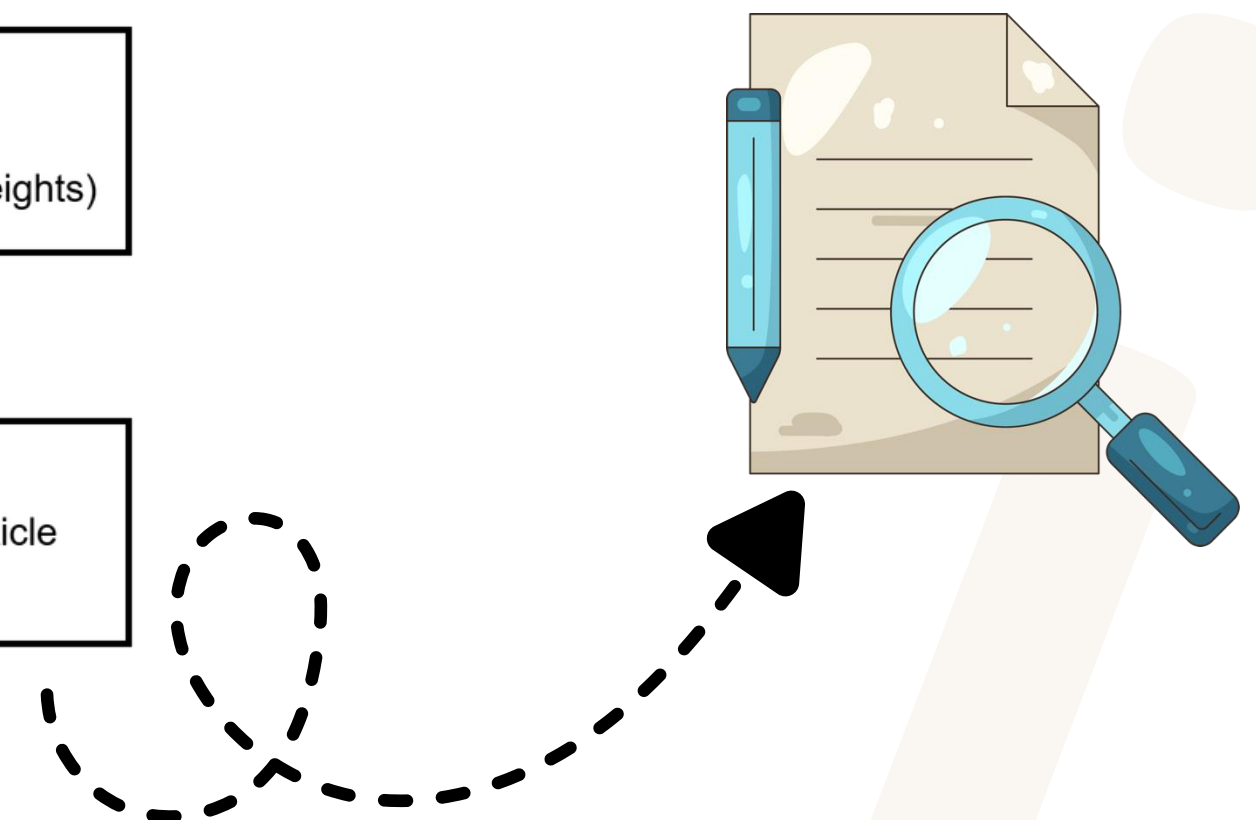
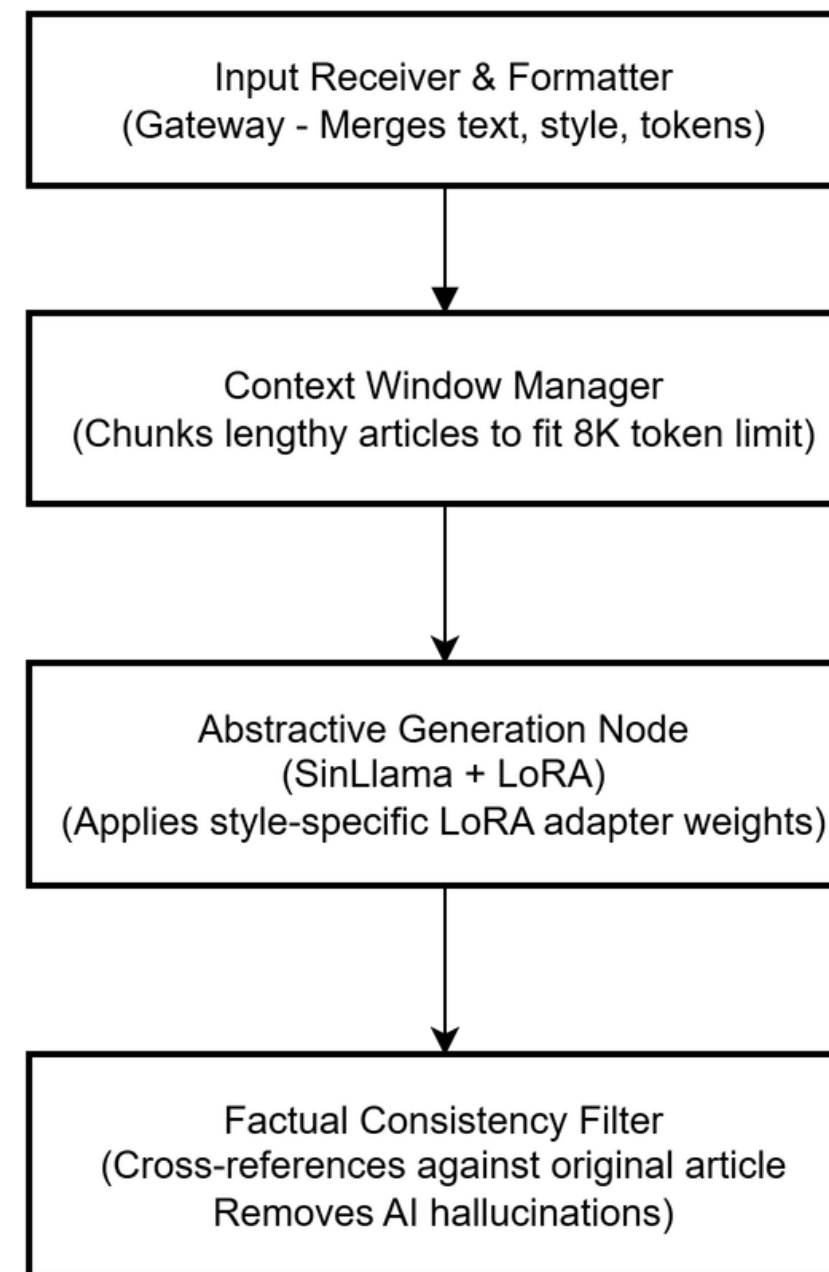
- Cross-checks with original article
- Removes hallucinated facts

Sinhala News

Summarizer

TEAM INTEGRATON

- Receives generated summary
- Corrects grammar & spelling
- Final output shown to user





Sinhala Journalistic Style Rewriter

PROBLEM

Sinhala news platforms do not have AI tools that can automatically rewrite the same news article into different journalistic styles such as editorial, sports, youth, or children's writing.

As a result, journalists must manually rewrite content, which is time-consuming and inconsistent.



Research Gap

Existing Sinhala NLP systems generate text in a single neutral writing style and lack mechanisms for explicit style control.

There is no validated Sinhala news rewriting system that can transform the same article into multiple journalistic styles

NOVELTY

A style-controlled Sinhala news rewriting system that uses a LoRA-fine-tuned Sinhala Large Language Model to transform the same news article into multiple journalistic writing styles while preserving semantic meaning and factual content.

OBJECTIVES

To develop a style-controlled Sinhala news rewriting system that can transform news articles into multiple journalistic styles without changing the original meaning.

Sinhala Journalistic Style Rewriter

Sub-Objectives

- Prepare a Sinhala news dataset suitable for style transformation tasks.
- Identify different news writing styles.
- Train the SinLlama model using LoRA.
- Use prompts to control the writing style.

Sinhala Journalistic Style Rewriter

Research Methodology:

Design Science Research approach focusing on building and experimentally evaluating a style-controlled text generation system.

Latest Technologies Used

- Fine-tuned Sinhala Large Language Model (SinLlama)
- LoRA (Low-Rank Adaptation) for efficient fine-tuning
- Transformer-based architecture
- Style-conditioned text generation using prompts

Data Source

- Data is collected from publicly available Sinhala news websites.

Sinhala Journalistic Style Rewriter

System Overview

- Input Sinhala news article
- Text preprocessing (cleaning and normalization)
- Style selection (formal, editorial, sports, youth, children)
- LoRA-adapted language model rewrites the article
- Output rewritten article in selected style

Integration with Overall System

The Style Rewriter works with other SinhalaJournal-LLM modules such as:

- Grammar Checker

Validation (during development / training)

Semantic preservation - Compare embeddings of original vs. rewritten text to ensure meaning is preserved.

Style learning - Use a pretrained or small style classifier to check if the rewritten text is going toward the target style.

Evaluation (after generation / final testing)

Style accuracy- Precision, Recall, F1-score

Human evaluation -
Native Sinhala speakers evaluate: meaning preservation, style correctness, readability, overall quality.

Commercialization Plan

TARGET MARKET

- News Media Organizations
- Digital News Platforms
- Journalism Schools / Media Training Institutes

REVENUE MODEL

- SaaS Subscription
- Enterprise Licensing

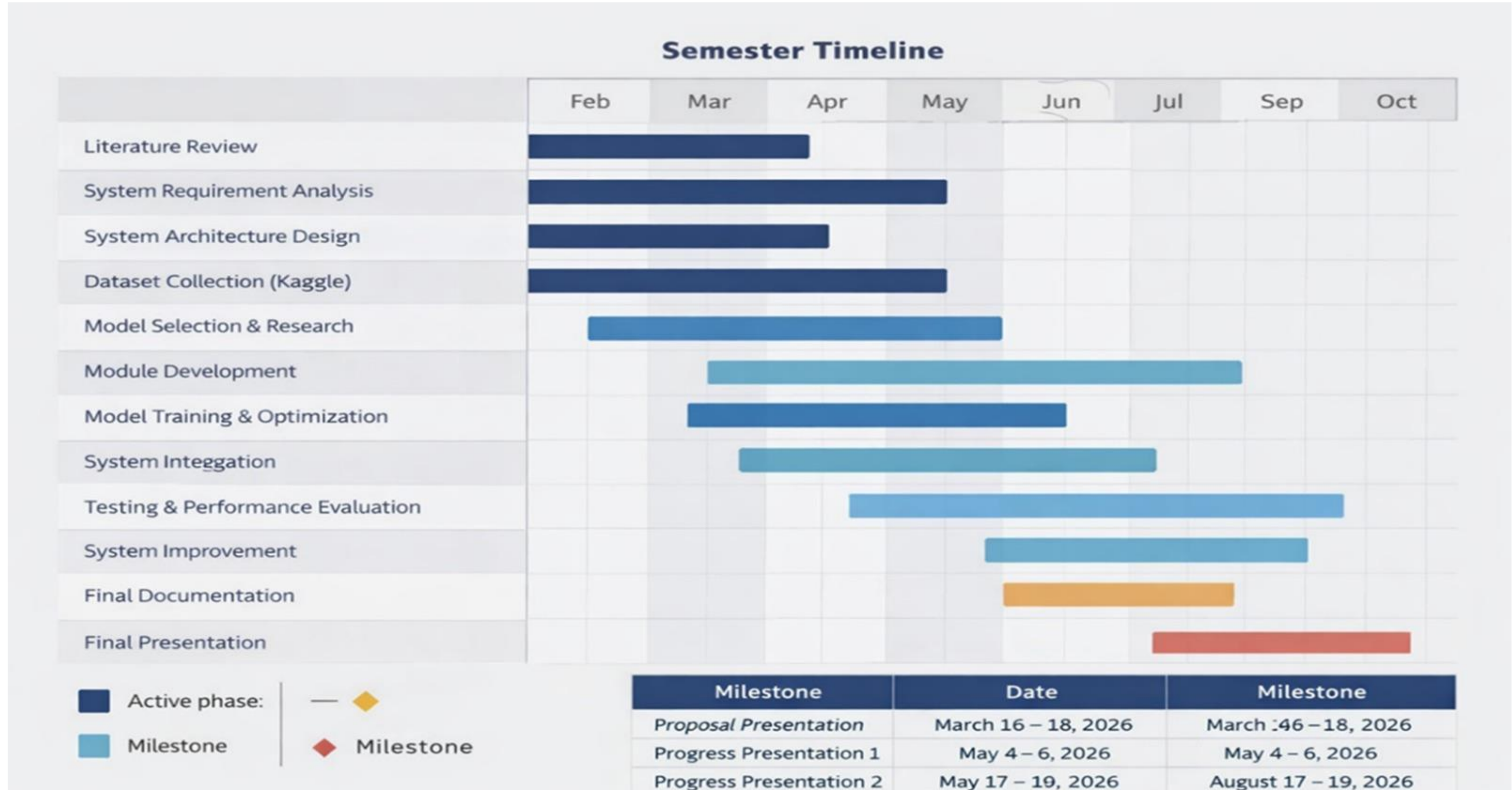
VALUE PROPOSITION

- Write summaries faster
- Reduce manual editing work
- Maintain consistent news style
- Publish news quicker

COST STRUCTURE

- Cloud servers (model hosting)
- GPU compute for AI inference
- System maintenance
- Developer & support team

Gantt Chart





Thank You